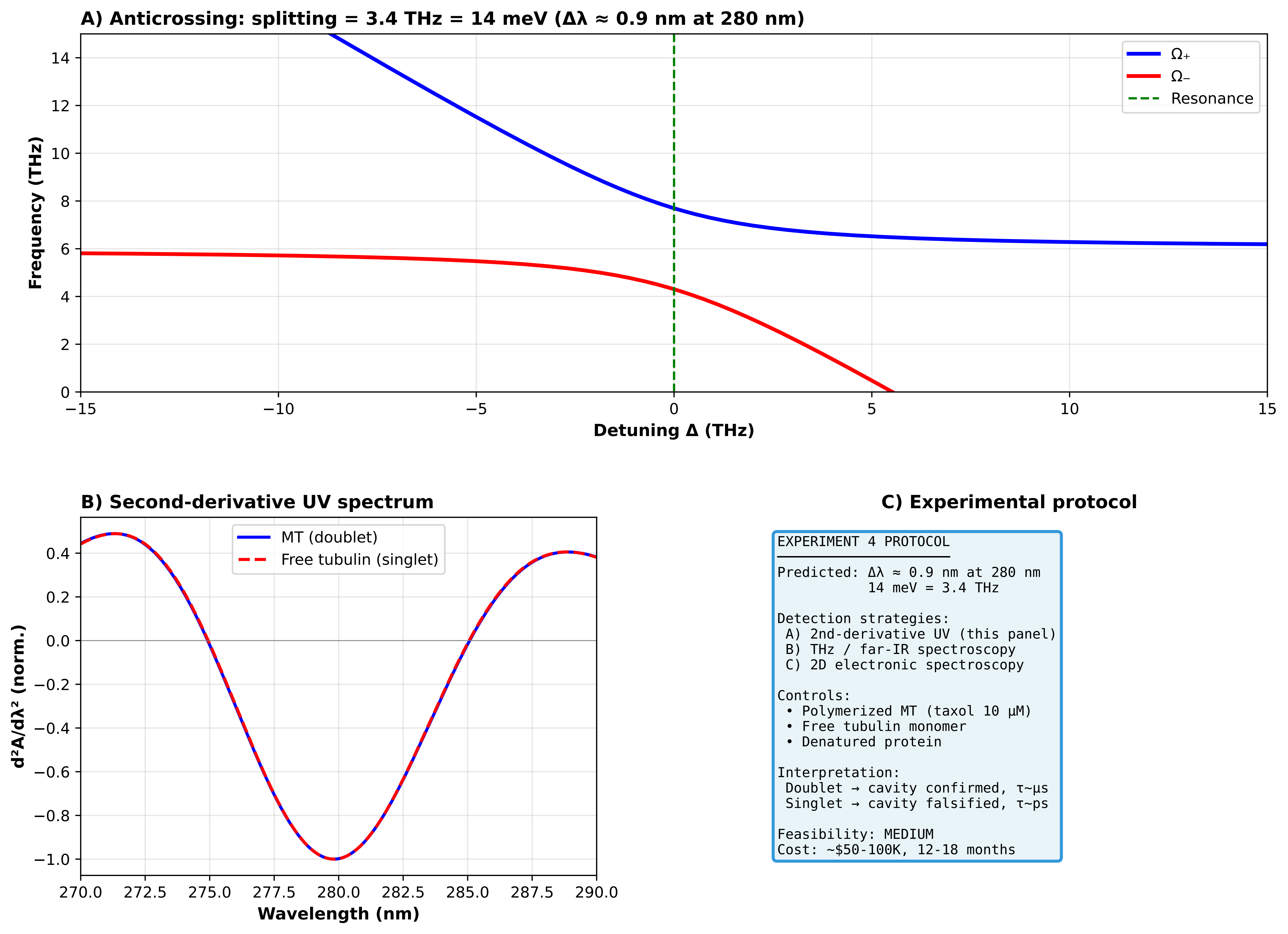
A) Anticrossing: splitting = 3.4 THz = 14 meV (Δλ ≈ 0.9 nm at 280 nm)
Ω+
Ω−
Resonance
Frequency (THz)
Detuning Δ (THz)
B) Second-derivative UV spectrum
MT (doublet)
Free tubulin (singlet)
d²A/dλ² (norm.)
Wavelength (nm)
C) Experimental protocol
EXPERIMENT 4 PROTOCOL
Predicted: Δλ ≈ 0.9 nm at 280 nm
14 meV = 3.4 THz
Detection strategies:
A) 2nd-derivative UV (this panel)
B) THz / far-IR spectroscopy
C) 2D electronic spectroscopy
Controls:
• Polymerized MT (taxol 10 μM)
• Free tubulin monomer
• Denatured protein
Interpretation:
Doublet → cavity confirmed, τ~μs
Singlet → cavity falsified, τ~ps
Feasibility: MEDIUM
Cost: ~$50-100K, 12-18 months